

Purple-
pink crystals

Cobalt

27
Co

Cobalt is used to make permanent magnets (magnets that don't need electricity to work).

Cobalt is important biologically—it is part of vitamin B12, which is important for making red blood cells. Alloys containing cobalt are hard-wearing and are used to make jet-engine blades and prosthetic body parts.



Erythrite

A compound of cobalt, arsenic, oxygen, and water, erythrite is a mineral ore of cobalt. Bright red-purple in appearance, it is called "cobalt bloom" by miners.



Irradiating food

One particular form (isotope) of cobalt, cobalt-60, emits gamma radiation. This radiation can be used to sterilize surgical instruments and to preserve food by killing harmful germs.

Coloring glass

In ancient times, a popular use of cobalt was in a pigment for coloring porcelain. In modern times, cobalt is often used to give glass a bright blue color.



This stained glass gets its blue color from cobalt.

Table lamp with blue glass lampshade

Nickel

28
Ni

This element's name comes from the German word *kupfernickel*, meaning "devil's copper."

At first, German miners thought this mineral contained copper. It was later found to be an ore of nickel, but the shortened name—nickel—stuck. Pure nickel does not rust and is used to coat metal objects to protect them from corrosion.



Pure nickel balls refined in a laboratory

These silvery-white pellets have a yellowish tinge.

Tagging sea turtles

Nickel alloys are used to make numbered tags that help scientists track and identify individual marine creatures, such as sea turtles. Known as Monel tags, these are mixtures of nickel and copper and are highly resistant to corrosion in water.



Copper nickel

The 5-cent coin called the "nickel" is in fact three-quarters copper and only one-quarter nickel. Nickel appears in most "silver" coinage in the form of cupronickel, the alloy that nickel forms with copper.

The George Washington quarter, also made of cupronickel, was created in 1932 to commemorate the 200th anniversary of the president's birth.

